Khanh-Tung Tran

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Summary

Data Scientist and PhD Researcher with 5 years of experience developing, validating, deploying, and monitoring machine-learning systems across time-series, speech, language, and production AI applications.

Strong background in Python, SQL, feature engineering, anomaly and performance analysis, model evaluation, and translating experimental results into clear technical and business recommendations.

Production experience spanning data pipelines, REST APIs, PostgreSQL, distributed GPU infrastructure, CI/CD, observability, and cross-functional delivery; motivated to apply this foundation to payment-fraud detection and prevention.

Education

PhD **University College Cork**, Computer Science — Cork, Ireland — Jan 2024 – Jan 2027
- Supervisor: Assoc. Prof. Harry Nguyen & Prof. Barry O'Sullivan.
- Thesis: Data-Efficient Cross-Lingual Understanding, Generation, and Alignment in Large Language Models.

BSc **Ho Chi Minh City University of Technology**, Computer Science — Ho Chi Minh City, Vietnam — Oct 2018 – Nov 2022
- Major: Computer Science. Honors Program. GPA: 9.33/10, top 5/400.
- Thesis: Development of an Expressive Neural Text-to-Speech System.

Work Experience

CloudCIX Ltd, AI Architect Consultant — Ireland — June 2024 – present
- Developed and deployed multi-tenant machine-learning systems and data-processing pipelines, converting new document and user data into production model signals.
- Built continuous model and knowledge-integration pipelines covering data preparation, evaluation, deployment, and operational feedback.
- Implemented CI/CD and post-deployment monitoring using Docker, Kubernetes, FastAPI, PostgreSQL, and OpenTelemetry on a multi-node GPU cluster with 16 NVIDIA B200 GPUs.
- Investigated latency and reliability issues using telemetry and profiling data, communicating findings and recommended improvements across technical teams.

Umeå University, Research Engineer — Sweden — Aug 2023 – Jan 2024
- Built and administered Slurm-based HPC infrastructure, Linux modules, Docker, and enroot containers.
- Deployed reproducible GPU environments supporting model training, inference, validation, job scheduling, monitoring, logging, and alerting.
- Engineered a 3D Digital Avatar with NVIDIA Omniverse and NVIDIA Riva, integrating facial animation, ASR, TTS, and conversational AI.

FPT Software, AI Researcher — Vietnam — May 2021 – July 2023
- Developed machine-learning models for speech, audio, and time-series detection tasks using Python, PyTorch, Transformers, torchaudio, librosa, Lightning, and Hydra.
- Owned experiments across data exploration, preprocessing, feature engineering, modelling, validation, hyperparameter tuning, and inference optimisation.
- Analysed model errors and performance metrics across experimental cohorts, using controlled comparisons to guide model and data improvements.

Projects

Guiden

Multi-tenant machine-learning system deployed on private-cloud infrastructure.

- Built data ingestion, embedding, retrieval, and model-serving workflows using Python, PostgreSQL, Kubernetes, FastAPI, nginx, and BentoML.
- Used profiling and monitoring evidence to isolate bottlenecks and reduce inference latency by 20%.
- Automated deployment through CI/CD and implemented authentication and SSO using LDAP.

ML-Workbench

Production data-processing, search, and retrieval platform.

- Engineered ingestion and feature-generation workflows for scraping, parsing, chunking, embedding, and indexing heterogeneous data streams.
- Implemented SQL-based full-text and vector similarity search with PostgreSQL tsvector and pgvector.
- Orchestrated durable workflows with Hatchet, including failure handling and repeatable execution.

Publications

Disentangling Continued Pre-training: Attention-Driven Routing and Semantic Hub Preservation in Language Adaptation — ACL 2026
Tran, K. T., Tran, V. K., O'Sullivan, B., Nguyen, H. D.

LaCoMSA: Language-Consistency Multilingual Self-Alignment with Latent Representation Rewarding — EACL 2026
Tran, K. T., O'Sullivan, B., Nguyen, H. D.

IRLBench: A Multi-modal, Culturally Grounded, Parallel Irish-English Benchmark for Open-Ended LLM Reasoning Evaluation — KDD 2026
Tran, K. T., Nguyen, D. H., O'Sullivan, B., Nguyen, H. D.

Reasoning Transfer for an Extremely Low-Resource and Endangered Language: Bridging Languages Through Sample-Efficient Language Understanding — AAAI-26
Tran, K. T., O'Sullivan, B., Nguyen, H. D.

Disentangling Language Understanding and Reasoning Structures in Cross-lingual Chain-of-Thought Prompting — EMNLP 2025
Tran, K. T., Vu, H., O'Sullivan, B., Nguyen, H. D.

UCCIX: Irish-eXcellence Large Language Model — ECAI 2024
Tran, K. T., O'Sullivan, B., Nguyen, H. D.

Achievements

- 30 Under 30 in Ireland 2026, The Business Post.
- Best Paper Award, LoResLM Workshop, EACL 2026.

Technical Skills

Machine Learning: Anomaly detection, supervised learning, decision trees, random forests, gradient-boosted trees, deep learning, time-series modelling, feature engineering, hyperparameter tuning, model validation, A/B testing, error analysis, post-live monitoring

Data & Programming: Python, SQL, Databricks, PostgreSQL (tsvector/pgvector), pandas, NumPy, PyTorch, TensorFlow, Transformers, C/C++, Bash, data exploration and preprocessing

Production ML: FastAPI, REST APIs, Kubernetes, Docker, CI/CD, vLLM, BentoML, Prometheus, Grafana, Elasticsearch, Sentry, OpenTelemetry, distributed and asynchronous systems

Graph & Workflow Tools: NetworkX, Hatchet, RabbitMQ, Celery; knowledge of graph-based relationship analysis and willingness to develop Neo4j or TigerGraph expertise